GPU LAB / PERFORMANCE & EFFICIENCY

GPU Lab - demonstration report

Synthetic preview of the interface and report format. No physical GPU measurements.

SIMULATED DEMONSTRATION - SYNTHETIC VALUES. NOT HARDWARE MEASUREMENTS.

Run	GPU / environment
792d5805-854d-4c64-bed2-b658e29e195d	Demonstration GPU - synthetic data
Status: completed	Profile: quick
Started: 2026-10-29T03:36:00.843+00:00	Suite: 0.1.0
Metadata revision: 1	Exported: 2026-10-29T03:36:51.234+00:00

Read this report

Results measure the selected GPU with its installed driver and libraries. GPU power only; whole-server power is excluded. Higher throughput and work per joule are better. Lower latency and energy per identical work unit are better. Power and temperature require context. There is no overall score.

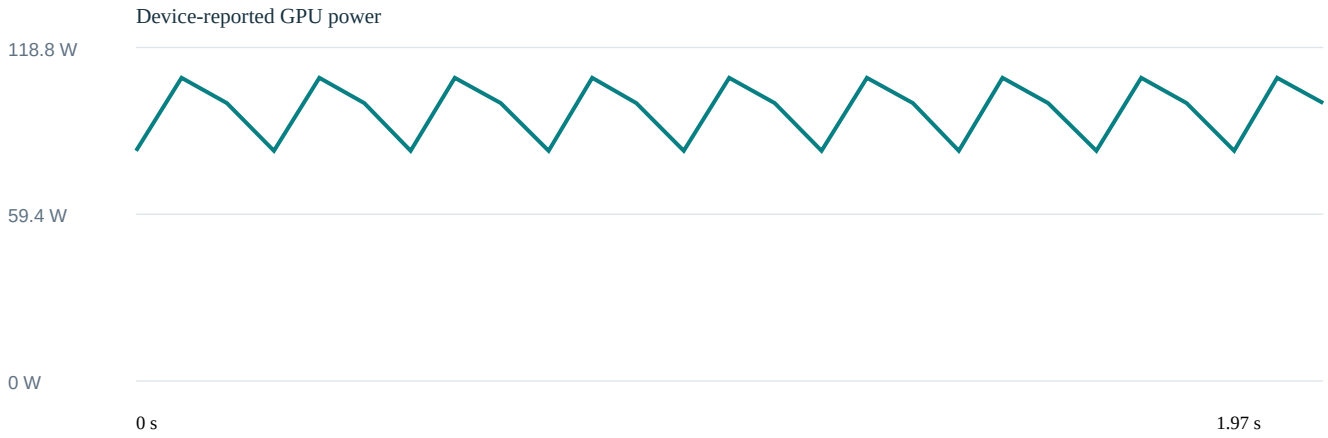
Coverage

Workload / representative case	Result	Status
General arithmetic / FP32	12,500 GFLOP/s / Higher is better	passed
Dense matrix math / FP32 / square	24.8 TFLOP/s / Higher is better	passed
VRAM & cache behavior / copy / 64 MiB	320 GB/s / Higher is better	passed
Host <-> GPU transfers / upload / 64 MiB	24.5 GB/s / Higher is better	passed
Small-job latency / host completion	32.4 us / Lower is better	passed
Hardware video / H264 / 1080p encode	440 frames/s / Higher is better	passed
Concurrent workloads / 2 concurrent jobs	16,800 GFLOP/s / Higher is better	passed
Sustained workloads / sustained FP32	12,300 GFLOP/s / Higher is better	passed
VRAM integrity / VRAM mismatches	0 errors / Lower is better	passed

Representative cases are selected by workload definition, not the largest score. Full results and precision/size details follow. Compare only matching configurations.

GPU power over time

Full-run telemetry includes warm-up, measured work and idle gaps. Each case uses its own covered interval for energy integration.



General arithmetic

Register-heavy FP32 FMA and INT32 arithmetic on general compute units.

Status: passed

Case / metric	Result	Direction
FP32 / throughput	12,500 GFLOP/s	Higher is better
average GPU power	100 W	Context - interpret with workload
work per joule	125 GFLOP/J	Higher is better
INT32 / throughput	8,100 GOP/s	Higher is better
average GPU power	100 W	Context - interpret with workload
work per joule	81 GOP/J	Higher is better

Methodology

Independent accumulators per work item; fixed iteration count; FP contraction controlled. Integer operations are modulo 2^32. Outputs are checked against host references. No matrix acceleration is requested.

Timing boundary: OpenCL event timing excludes compilation and allocation; efficiency uses continuous-loop wall time.

FP32

Status: passed

SIMULATED: not a GPU measurement

Parameter	Value
fixture	synthetic-v1
profile	quick

INT32

Status: passed

SIMULATED: not a GPU measurement

Parameter	Value
fixture	synthetic-v1
profile	quick

Dense matrix math

FP32, FP16, BF16 and INT8 GEMM with correctness checks for each precision.

Status: passed

Case / metric	Result	Direction
FP32 / square / throughput	24.8 TFLOP/s	Higher is better
average GPU power	100 W	Context - interpret with workload
work per joule	0.25 TFLOP/J	Higher is better
FP16 / square / throughput	102.4 TFLOP/s	Higher is better
average GPU power	100 W	Context - interpret with workload
work per joule	1.02 TFLOP/J	Higher is better
INT8 / square / throughput	190 TOP/s	Higher is better
average GPU power	100 W	Context - interpret with workload
work per joule	1.9 TOP/J	Higher is better

Methodology

Vendor PyTorch matmul on fixed seeded matrices. TF32 disabled, FP16/BF16 reduced-precision reductions disabled where exposed; accumulation policy recorded. INT8 uses the probed private torch._int_mm API and INT32 output; unavailable paths are skipped. Dense operations count a multiply and add as two operations. Library dispatch is not proof of a specific hardware instruction.

Timing boundary: GPU event time excludes input creation/transfers. Sustained-loop wall throughput is used for efficiency.

FP32 / square

Status: passed

SIMULATED: not a GPU measurement

Parameter	Value
fixture	synthetic-v1
profile	quick

FP16 / square

Status: passed

SIMULATED: not a GPU measurement

Parameter	Value
fixture	synthetic-v1
profile	quick

INT8 / square

Status: passed

SIMULATED: not a GPU measurement

Parameter	Value
fixture	synthetic-v1
profile	quick

VRAM & cache behavior

Read, write, copy and strided throughput plus dependent pointer-chase latency.

Status: passed

Case / metric	Result	Direction
copy / 64 MiB / throughput	320 GB/s	Higher is better
average GPU power	100 W	Context - interpret with workload
work per joule	3.2 GB/J	Higher is better
dependent access / latency	180 ns/access	Lower is better

Methodology

Working sets sweep from cache-sized to large allocations. Read/write count one byte stream; copy counts read plus write. Pointer chase is a serial dependency chain through a seeded permutation; reported latency includes instruction overhead and is not a pure DRAM latency measurement.

Timing boundary: GPU event timing. Cache residency is inferred from size, not asserted from hardware counters.

copy / 64 MiB

Status: passed

SIMULATED: not a GPU measurement

Parameter	Value
fixture	synthetic-v1
profile	quick

dependent access

Status: passed

SIMULATED: not a GPU measurement

Parameter	Value
fixture	synthetic-v1
profile	quick

Host <-> GPU transfers

Pinned host-memory upload, download and simultaneous-direction transfer.

Status: passed

Case / metric	Result	Direction
upload / 64 MiB / throughput	24.5 GB/s	Higher is better
average GPU power	100 W	Context - interpret with workload
work per joule	0.24 GB/J	Higher is better
download / 64 MiB / throughput	23.9 GB/s	Higher is better
average GPU power	100 W	Context - interpret with workload
work per joule	0.24 GB/J	Higher is better

Methodology

ALLOC_HOST_PTR mapped buffers are filled before timing. Message sizes sweep from 4 KiB to the configured buffer limit. Separate profiling queues measure concurrent directions. Reported duplex throughput is the sum of both directions over their combined span.

Timing boundary: Event transfer time and wall completion latency; host allocation and pinning excluded. Host/NUMA/PCIe affect these system-level results.

upload / 64 MiB

Status: passed

SIMULATED: not a GPU measurement

Parameter	Value
fixture	synthetic-v1
profile	quick

download / 64 MiB

Status: passed

SIMULATED: not a GPU measurement

Parameter	Value
fixture	synthetic-v1
profile	quick

Small-job latency

Submission, device execution and round-trip completion overhead.

Status: passed

Case / metric	Result	Direction
host completion / latency	32.4 us	Lower is better

Methodology

Dispatch a one-work-item kernel and wait for it repeatedly. Measure host enqueue duration, GPU event duration, and host completion duration independently. Report median and p95; this is a driver/runtime/system measurement.

Timing boundary: Compilation and warm-up excluded. Individual host timings include Python wrapper overhead.

host completion

Status: passed

SIMULATED: not a GPU measurement

Parameter	Value
fixture	synthetic-v1
profile	quick

Hardware video

H.264, HEVC and AV1 hardware encode, decode and transcode, with quality analysis.

Status: passed

Case / metric	Result	Direction
H264 / 1080p encode / throughput	440 frames/s	Higher is better
average GPU power	100 W	Context - interpret with workload
work per joule	4.4 frames/J	Higher is better
VMAF	94.2 score	Higher is better
output bitrate	8 Mbit/s	Context - interpret with workload
HEVC / 4K encode / throughput	110 frames/s	Higher is better
average GPU power	100 W	Context - interpret with workload
work per joule	1.1 frames/J	Higher is better
VMAF	94.2 score	Higher is better
output bitrate	8 Mbit/s	Context - interpret with workload

Methodology

Use explicitly selected NVENC/NVDEC or VAAPI devices. Fixed seeded moving-detail test clip or user-supplied reference. Same resolution, bit rate, frame rate, GOP and pixel format. Encode quality is checked after timing using PSNR/SSIM and VMAF if available. Presets are recorded, not assumed equivalent across vendors. Decode uses the generated hardware-encoded stream; this input dependency is recorded.

Timing boundary: Process wall time includes startup and encode uploads. Decode readback is included; transcode retains hardware frames. CPU reference preparation and quality analysis excluded. FPS is throughput, not per-frame latency.

H264 / 1080p encode

Status: passed

SIMULATED: not a GPU measurement

Parameter	Value
fixture	synthetic-v1
profile	quick

HEVC / 4K encode

Status: passed

SIMULATED: not a GPU measurement

Parameter	Value
fixture	synthetic-v1
profile	quick

Concurrent workloads

Compute queues, hardware video streams and mixed compute/video sharing one GPU.

Status: passed

Case / metric	Result	Direction
2 concurrent jobs / throughput	16,800 GFLOP/s	Higher is better
average GPU power	100 W	Context - interpret with workload
work per joule	168 GFLOP/J	Higher is better

Methodology

Identical arithmetic jobs with distinct outputs run through independent OpenCL queues and host threads. Media cases use one, two and four hardware encoders; a mixed case adds one arithmetic loop. Video input preparation and process startup are recorded boundaries. Report aggregate throughput and per-job slowdown against the single-job baseline.

Timing boundary: Wall time from submitting concurrent jobs until all complete, including host scheduling.

2 concurrent jobs

Status: passed

SIMULATED: not a GPU measurement

Parameter	Value
fixture	synthetic-v1
profile	quick

Sustained workloads

Compute, memory and hardware video phases with time-series throughput.

Status: passed

Case / metric	Result	Direction
sustained FP32 / throughput	12,300 GFLOP/s	Higher is better
average GPU power	100 W	Context - interpret with workload
work per joule	123 GFLOP/J	Higher is better

Methodology

Split the configured total soak duration across selected compute, memory and video phases. Check compute/memory outputs, record throughput and telemetry, compare early and late intervals. Video uses the first configured codec, resolution and mode, with preparation excluded and encode process boundaries included.

Timing boundary: Wall time per interval, including checks; GPU power collected independently. User temperature ceiling stops the run if exposed sensor reaches it.

sustained FP32

Status: passed

SIMULATED: not a GPU measurement

Parameter	Value
fixture	synthetic-v1
profile	quick

VRAM integrity

Address-dependent pattern verification across a configurable memory allocation.

Status: passed

Case / metric	Result	Direction
VRAM mismatches / mismatches	0 errors	Lower is better

Methodology

Allocate bounded VRAM; write zero, all-one, alternating and address-hashed patterns in multiple passes, then GPU-verify every word and read mismatch counts. Host spot-checks verify the verifier. Cannot allocate driver-reserved memory; this is not a hardware certification or an ECC scrub.

Timing boundary: Reports tested MiB, patterns, passes, mismatch counts and elapsed time. Other workloads' correctness checks remain enabled.

VRAM mismatches

Status: passed

SIMULATED: not a GPU measurement

Parameter	Value
fixture	synthetic-v1
profile	quick

Configuration & provenance

Current editable metadata is separated from original detected evidence. Changes do not rewrite measured results.

Current metadata

Field	Value
name	GPU Lab - demonstration report
description	Synthetic preview of the interface and report format. No physical GPU measurements.
tags	[]
notes	
hardware.host.hostname	demonstration
hardware.host.os	simulated
hardware.host.cpu	simulated
hardware.gpu.id	demo:0
hardware.gpu.name	Demonstration GPU - synthetic data
hardware.gpu.vendor	demo
hardware.gpu.simulated	True
hardware.gpu.vram_bytes	25769803776

Original detected configuration

Field	Value
host.hostname	demonstration
host.os	simulated
host.cpu	simulated
gpu.id	demo:0
gpu.name	Demonstration GPU - synthetic data
gpu.vendor	demo
gpu.simulated	True
gpu.vram_bytes	25769803776
app_version	0.1.0
captured	2026-10-29T03:36:00.842+00:00

Test configuration

Field	Value
duration	2.0
warmup	0.5
repeats	3
buffer_mib	64
matrix_sizes	[1024, 2048]
soak_seconds	30
video_seconds	3
profile	quick
sample_interval	1.0
precisions	["fp32", "fp16", "bf16", "int8"]
codecs	["h264", "hevc", "av1"]
resolutions	["1920x1080", "3840x2160"]
video_modes	["quality", "low-latency"]
reference	None
max_temp	90.0
memory_fraction	0.5
integrity_passes	2
timeout	7200
concurrency_media	True
soak_workloads	["compute", "memory", "video"]
allow_partial	False

Metadata history

2026-10-29T03:36:00.844+00:00 - created
2026-10-29T03:36:00.878+00:00 - preflight